

Mating Mechanism Operational Specification: Decomposing B1.10 by Formalizing Cross-Layer Combination of DNA from Different Sources Through Three Patterns That Collapse Biology's Separate Inheritance Channels into One Governed Mechanism

A derivation note from the Coordination Knowledge Substrate (CKS) theory series.

Author: Wenxin Li (Independent Researcher)

ORCID: 0009-0004-8065-3235

Date: May 12, 2026

License: Creative Commons Attribution 4.0 International (CC BY 4.0)

Attribution

This work derives from and formalizes the instinct/reasoning separation pattern introduced in *The Instinct/Reasoning Separation Outside the Model* (Li, April 2026), the second paper in the CKS theory series following *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026). It does not introduce new axioms. Its sole contribution is to formalize, as an independently citable operational specification, the mating mechanism that Paper 2 introduces as a governed lifecycle primitive — articulating precisely what mating is architecturally, how it operates across DNA and Action layers, how it collapses biology's separate inheritance channels into one mechanism, and what governance properties it inherits from the CKS pattern.

Abstract

Paper 2 names mating as one of three lifecycle primitives (birth, mating, death) that apply uniformly at every level of a CKS Self — cell, aspect, and Self. This note formalizes mating as a cross-layer combination mechanism: the architectural primitive by which DNA from different sources is combined into offspring DNA through one of three patterns (union, selective merge, lineage-preserved union), under human governance, to produce an offspring specification that enters birth. The note articulates the mechanism precisely, identifies what makes it architecturally distinctive, traces the biological analog and CKS advantage, states the inherited Paper 1 commitments the mechanism carries forward, describes operational implications for deployment configurations, and names the limits that bound the mechanism's scope. Mating's defining architectural contribution is collapsing what biology distributes across multiple inheritance channels — genetic, epigenetic, behavioral, and cultural transmission — into one governable combination primitive. B2.45 is the first of six notes decomposing B1.10; subsequent notes (B2.46–B2.50) formalize each pattern and the governance and verification dimensions in detail.

1. Why mating mechanism operational specification needs to be formalized as a standalone note

B1.10 names mating as the governed lifecycle primitive for combining DNA from different sources into offspring DNA. The foundational commitment is stated compactly in Paper 2 (§6.3): three pattern variants exist, the mechanism collapses biology's separate inheritance channels, and the operation is governed by orchestration substrate. What B1.10 does not do, by design, is elaborate each pattern's operational properties in isolation, or distinguish the mechanism's cross-layer character as a standalone architectural claim. Phase B2's structural logic is operational-variant-as-architectural-decomposition: where a Phase B1 note states a commitment, Phase B2 notes extract and formalize each operationally distinct derivation that commitment supports.

B2.45 opens that decomposition at the mechanism level — before the individual patterns are elaborated. The strategic purpose of extracting the mechanism specification as its own note is that the mechanism is itself a patentable derivation distinct from any single pattern. A claim that mating can draw from the action layer as well as the DNA layer, that action-layer-derived content can flow into offspring DNA, and that all combination operates through one governed primitive rather than through separate machinery for genetic versus cultural inheritance — this is an architectural commitment that holds regardless of which of the three patterns is applied. Formalizing it separately means it stands as prior art independent of the pattern-level elaborations in B2.46–B2.48.

B2.45 is the forty-fifth Phase B2 note, opening the B1.10 mating decomposition following the completion of the B1.09 birth decomposition (B2.40–B2.44). Six notes compose this decomposition: B2.45 (this note, mechanism specification), B2.46 (union pattern), B2.47 (selective merge pattern), B2.48 (lineage-preserved union pattern), B2.49 (mating governance and lineage establishment), B2.50 (mating verification). The death decomposition begins at B2.51.

2. The architectural mechanism precisely stated

Mating in CKS is the cross-layer combination of DNA from different sources into offspring DNA, governed by orchestration substrate, through one of three patterns, producing an offspring specification that enters birth.

Each component of this definition carries architectural weight.

Cross-layer combination. Mating is cross-layer because it can draw from more than one substrate layer within the CKS cell architecture. A cell carries two layers per B1.03: the DNA layer (stabilized orchestration rules, behavioral substrates, accumulated specifications) and the action layer (recorded task instances and outputs from prior execution). Mating can combine DNA-layer content from one cell with DNA-layer content from another cell — the straightforward genetic-analog case. It can also combine DNA-layer content with content derived from a cell's action layer: patterns distilled from action-layer accumulation can be promoted into DNA-like specifications and incorporated into offspring DNA through mating. This second capability — action-layer evidence flowing into offspring via mating — is what makes the mechanism genuinely cross-layer rather than merely cross-cell.

From different sources. The sources in a mating event are the parents. Parents may be two cells mating to produce offspring combining their DNA. They may be a cell and its own action-layer distillation, producing offspring that incorporates learned patterns. They may be cells from different aspects or different Selves, producing offspring that crosses structural boundaries. CKS does not restrict mating to two-parent biological-style configurations; the architecture commits to governed cross-source combination, and whether a specific instance is two-parent, self-derived, or multi-source is configured by orchestration substrate governance.

Into offspring DNA. The output of mating is a specification — offspring DNA — ready for birth per B1.09. The mating mechanism produces a specification; birth produces an entity from the specification. The offspring is a new entity, not a modified version of either parent. Birth verification per B2.44 applies to mating-derived births as to origination births.

Through three patterns. The three patterns configure how combination is performed:

Union: All DNA elements from both parent sources are included in the offspring. Merge-time conflicts — elements that cannot be trivially unified — are preserved as first-class substrate state in the offspring rather than resolved at merge time. This inherits Paper 1's conflict-preservation commitment (Claim 3): contradictions become persistent, addressable substrate content with provenance attached, not transient markers to be cleared before the merge completes.

Selective merge: Humans or LLMs operating under human direction pre-curate which elements from each parent source cross into the offspring. The curation rules are themselves substrate content authored under A2.04 governance, meaning the rules are human-authored, inspectable, modifiable, and overridable. Selective merge minimizes the offspring's content footprint while maximizing curation precision.

Lineage-preserved union: Performs the same full combination as union but additionally embeds pointers to parent cells in the offspring's birth record per A2.40's six provenance metadata fields. Every element of the offspring's substrate has traceable derivation back to its source parent, making the offspring's lineage substrate-addressable rather than merely documentable externally.

Governed by orchestration substrate. Humans authorize mating events; labor (including LLMs per A1.12) may perform combination operations. The governance-versus-labor distinction from A1.01 holds: who decides what to mate and through which pattern is a governance question; who performs the combination operation is a labor question. These are separately configured.

3. What makes mating mechanism specification architecturally distinctive

The mating mechanism's architectural distinctiveness rests on three properties that, taken together, do not appear in any single existing architecture.

First: biology's separate channels collapsed into one. Biology distributes inheritance across multiple analytically distinct channels. The Extended Evolutionary Synthesis position (Jablonka and Lamb; Laland and Uller) identifies at least four: genetic transmission (high-fidelity DNA copying), epigenetic transmission (methylation and chromatin state passing non-DNA information across cell divisions),

behavioral transmission (learned behaviors passed through social learning), and cultural transmission (symbolic and artifact-mediated inheritance). These channels operate through different molecular mechanisms, at different fidelities, and have been treated as a multi-layer inheritance architecture each with its own dynamics. CKS collapses all of them into one configurable combination primitive. The distinction between genetic-analog and cultural-analog inheritance does not appear as a mechanistic distinction in CKS; it appears as a configuration choice within the same governed mating mechanism.

Second: learned behaviors are heritable through mating. Because mating draws from the action layer as well as the DNA layer, patterns that accumulated through prior execution — behaviors that emerged from what a cell actually did rather than what it was designed to do — can propagate to offspring. A cell's action-layer content can be distilled into DNA-like specifications and incorporated into offspring DNA through mating. This means operational learning is architecturally heritable: what a deployment learned during operation can be carried forward into new entities through a governed combination event. Conventional AI deployment architectures have no mechanism for this. Components are created from scratch or copied; there is no architectural primitive that takes accumulated execution evidence and incorporates it into offspring specifications through a governed combination event.

Third: explicit governance over what combination produces what offspring. The three patterns provide governance-accessible control over the combination operation's outcome. The selection of pattern is itself a governance decision under A1.01. This is different from both biological inheritance (where the combination mechanism is not governed; recombination follows probabilistic molecular dynamics) and software merge operations (where merge rules are engineering-defined, not governance-defined). In CKS, the choice of which pattern to apply, which selection rules govern selective merge, and which parent lineage to preserve is substrate content under human authority.

Conventional AI architectures, including multi-agent frameworks and component-composition architectures, have no mating as an architectural primitive. Where combination of component content occurs, it is application-level merge logic sitting atop version-control infrastructure — not a governed lifecycle primitive with explicit pattern variants and cross-layer character.

4. The biological analog and CKS advantage

The biological analog illuminates both the mechanism's conceptual origin and the architectural advantage CKS achieves over it.

Sexual reproduction in biology combines DNA from two parents through meiosis and fertilization. The resulting offspring inherits from both parents through recombination events that produce novel genetic combinations not present in either parent. Cultural transmission passes learned behaviors across generations through social learning and symbol use, but through entirely different mechanisms — no genetic recombination is involved. Biology thus requires two separate systems to achieve what CKS handles through one primitive: genetic recombination for heritable content, cultural transmission for learned behaviors.

The CKS analog is richer in four respects. First, the combination primitive is governed: humans authorize what gets combined and through which pattern, whereas biological recombination follows molecular dynamics under no external governance. Second, the mechanism is configurable: the three patterns offer distinct architectural outcomes (conflict preservation, content minimization, full provenance), whereas biological recombination produces one class of outcome. Third, cross-lineage combination is available as architectural commitment rather than exception: CKS permits governance-configured combination across any cell types, whereas biology's general case in canonical multicellular eukaryotes is reproductive isolation making cross-species recombination rare and often maladaptive. Fourth, the mechanism is unified across inheritance types: DNA-layer and action-layer content combine through the same governed primitive, whereas biology requires separate mechanisms for genetic and cultural inheritance.

The analog functions as a conceptual scaffold: it names why mating is a fundamental lifecycle primitive worth naming, and it makes the inheritance intuition precise. The architectural substance is the unified governed combination mechanism, not the biological metaphor.

5. Inherited Paper 1 commitments

The mating mechanism inherits six Paper 1 commitments directly, and each imposes specific requirements on how mating operates.

A1.01 (human-governed — authority not labor). Mating is governed per A1.01. Humans hold the authority to authorize mating events, select patterns, author selection rules, and override any combination operation. Labor — the work of performing combination — is allocable to humans directly or to LLMs operating under human direction. The governance authority is not delegated; the combination labor is.

A1.12 (labor allocation framework — three modes). Combination operations in mating may be performed by humans directly (for high-stakes or novel combination events), by LLMs operating under human-authored rules (for routine combination at volume), or by stable cells executing under orchestration rules (for automated combination workflows). Which labor mode applies is a deployment-configuration choice under orchestration substrate governance.

A2.04 (rule authoring). Selection rules for the selective-merge pattern, and any orchestration rules governing which cells are eligible to mate, are substrate content authored by humans under A2.04's requirements. These rules are themselves substrate-addressable, modifiable, and subject to the three rights (inspect, modify, override) from A1.01.

A2.40 (six provenance metadata fields). Mating events are recorded with provenance. The six provenance metadata fields apply to the mating event itself (source parents, combination pattern applied, combination rules invoked, timestamp, authorized agent) and chain to the offspring's birth record per B2.43. For lineage-preserved union, the provenance record additionally embeds parent lineage pointers in the offspring's substrate, making every element traceable to its parent source.

A1.07 (path retraceability and accountability vocabulary). Offspring lineage in a mating event traces through the mating record to the parent sources. For lineage-preserved union, this traceability is embedded in the offspring's substrate. For union and selective merge, it is carried in the mating event's provenance record. The accountability vocabulary — which decision was made, by whom, under which rules, at which time — applies to mating events as to all governed operations.

A1.13 (composition requirements — no primitives bypassed). Offspring produced through mating must satisfy composition requirements. Birth verification per B2.44 applies to mating-derived births. An offspring specification produced by mating does not bypass architectural validation; it enters the same birth pathway as an origination-birth specification. Mating produces the specification; birth creates the entity and verifies it.

6. Operational implications

Deployments configure mating per their operational requirements. This section names the primary configuration dimensions and operational use cases without prescribing implementation.

Which cells can mate. Orchestration substrate governance determines which cell types, aspects, or Selves are eligible to mate with which others. Cross-aspect mating produces offspring combining patterns from distinct aspect domains. Cross-Self mating requires cross-Self authority under governance. Same-cell self-mating with action-layer distillation produces offspring incorporating learned behaviors from prior execution.

Which patterns apply to which events. The choice of union, selective merge, or lineage-preserved union is a governance decision configured per event type or per cell category. Events where conflict preservation matters most use union. Events where content minimization matters most use selective merge. Events where provenance must be substrate-embedded rather than externally documented use lineage-preserved union.

Creating specialized cells from existing cell knowledge. Mating is the mechanism for producing specialized cells that inherit capabilities from multiple parent sources. Where a deployment needs a cell that combines the behavioral patterns of two existing cells, mating — rather than from-scratch origination birth — is the appropriate mechanism. The offspring inherits the combined DNA of its parents per the selected pattern.

Combining learned behaviors into new entities. Where a cell's action-layer accumulation has produced patterns worth carrying forward, those patterns can be distilled into DNA-like specifications and incorporated into offspring DNA through mating. This is the operational pathway by which action-feedback evolution can cross entity boundaries: learned behaviors can propagate not only by modifying the parent's own DNA but by contributing to an offspring's DNA through a governed mating event.

Mating events as governance events. Each mating event is a governance event under A1.01. The authorization, pattern selection, source identification, and combination output are all logged with provenance per A2.40. Mating-derived births run birth verification per B2.44. The mating record chains

to the offspring's birth record per B2.43, establishing a lineage that traces the offspring's entire origination: which parents contributed, through which pattern, under which governance authorization.

7. Limits

Several limits bound the mating mechanism's scope and prevent misreading.

Mating does not mean uncontrolled combination. Mating is governed per A1.01. No combination event occurs outside governance authorization. An LLM performing a combination operation does so under human-authored rules that specify eligible sources, applicable patterns, and required verification.

Mating does not produce offspring that bypass birth verification. An offspring specification produced by mating is not an entity; it is a specification ready for birth. Birth per B1.09 — including verification per B2.44 — applies. Mating produces the input to birth, not the output.

Mating does not produce organisms in the biological sense. The offspring of a mating event is an entity specification — a DNA-layer substrate ready to instantiate a cell, aspect, or Self. The biological vocabulary (parents, offspring, inheritance) is a conceptual scaffold, not a claim that biological reproduction is occurring.

The three patterns are not exhaustive of all possible combination logics. Union, selective merge, and lineage-preserved union are the patterns Paper 2 names. Additional patterns may be authored per A2.04's rule-authoring mechanism if a deployment requires combination logic not covered by the three. The three patterns cover the conceptually significant space Paper 2 commits to; extensions are possible within the governance framework.

Mating does not replace origination birth. Origination birth per B1.09 — creating a new entity from scratch under human governance — remains available. Mating is the mechanism for combination-derived origination; it supplements but does not displace from-scratch creation.

Mating is not automatic. No mating event occurs autonomously. Governance authorization is required. The decision to mate, the selection of sources, and the selection of pattern are governance decisions, not automated lifecycle dynamics.

Mating does not operate between incompatible DNA sources. Combination rules authored per A2.04 specify which sources are valid combination partners and which pattern variants apply to which source combinations. A mating event between sources whose combination would violate authored rules is not a valid mating event; governance authorization does not extend to rule-violating combinations.

8. Architectural mechanism — one-sentence test

A CKS deployment instantiates the mating mechanism commitment if and only if: when a new entity specification is produced by combining DNA from two or more sources — whether from the DNA layers of existing cells, from action-layer-derived content, or from both — the combination is performed through one of the three named patterns (union, selective merge, or lineage-preserved union), authorized by human governance per A1.01, recorded with provenance per A2.40, and the

resulting offspring specification enters birth per B1.09 with verification per B2.44.

9. Why naming this specification as standalone matters; opening the B1.10 decomposition

The mating mechanism specification is not identical to any single pattern. A claim that CKS unifies biology's separate inheritance channels into one governed primitive, that action-layer content is heritable through mating, and that the mechanism is cross-layer rather than merely cross-cell — these hold regardless of which pattern is applied. Formalizing this as a standalone note establishes the unifying mechanism as prior art independent of the pattern-level elaborations.

The B1.10 decomposition proceeds across six notes. B2.45 (this note) establishes the mechanism. B2.46 will formalize the union pattern in full: the conflict-preservation inheritance from Paper 1's Claim 3, the persistence of merge-time conflicts as first-class substrate state, and the distinction from version-control merge operations that treat conflicts as transient. B2.47 will formalize selective merge: the governance-driven curation of combination content, the authored selection rules, and the distinction from software product-line selective composition. B2.48 will formalize lineage-preserved union: substrate-embedded provenance, parent lineage pointers in offspring birth records, and the distinction from external provenance annotation. B2.49 will formalize mating governance and lineage establishment: the authority-versus-labor structure for mating events, the cross-lineage governance requirements, and how mating creates lineage relationships through birth records. B2.50 will formalize mating verification: how birth verification per B2.44 applies specifically to mating-derived births, what verification must establish for mating-derived offspring, and how verification chains to mating provenance records.

After B2.50, Phase B2 continues with B2.51–B2.55 decomposing B1.11 death: functional obsolescence and lineage supersession as distinct architectural operations with distinct governance and provenance requirements.

The progression through Phase B2 systematically closes the territory where each governed lifecycle primitive could be claimed as novel invention without encountering prior art. B2.45 establishes that territory for the mating mechanism itself; the five notes that follow establish it for each pattern and governance dimension in turn.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

How to cite this note

Li, W. (2026). *Mating Mechanism Operational Specification: Decomposing B1.10 by Formalizing Cross-Layer Combination of DNA from Different Sources Through Three Patterns That Collapse Biology's Separate Inheritance Channels into One Governed Mechanism*. May 12, 2026. ORCID: 0009-0004-8065-3235.